

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ ASSISTANT API
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>React.js, HTML5, CSS3, JavaScript</i>	<i>Responsive design, component reusability, fast UI for chat interface</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Node.js + Express.js</i>	<i>High scalability, non-blocking I/O, perfect for REST APIs /api/faqs</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>MongoDB + Mongoose</i>	<i>NoSQL flexible schema, easy to store FAQs as JSON, fast search</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render / Vercel + Docker</i>	<i>Easy scalability, free hosting for students, zero-config deployment for Node.js. Docker ensures app runs same in all environments</i>
5	Version Control & CI/CD	<i>GitHub + GitHub Actions</i>	<i>Seamless team collaboration, version tracking, automated testing & deployment on every push to main branch</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Google Gemini API + dotenv + Mongoose</i>	<i>Gemini API enables intelligent FAQ answering (core feature). dotenv secures API keys. Mongoose connects Node.js with MongoDB easily</i>